Memo: Cutting COD returns without cutting growth

To: Founder From: Kaveri Dalvi Re: COD decision engine (RTO Guard)

Bottom line. Shipping a COD order is still profitable up to roughly 70% return risk on an average ₹649 order, because a return costs us ₹169 while a delivered order earns ₹402. So cancelling "risky" orders is the most expensive way to fight RTO. The better tools are a cheap WhatsApp confirmation and a prepaid discount. On September 2026, the most recent month in the data (329 sample orders), these rules would have added an estimated net ₹6,864. Scaled to our real COD volume (about 1,650 orders a month), that is roughly ₹34,000 a month under my base assumptions, and about ₹6,000 a month under pessimistic ones (section 4). These are estimates, so we should measure the assumptions before relying on the larger figure.

1. What surprised me in the data (1,984 clean orders, RTO 28.5%)

- 1. Seven cities drive the problem. Gorakhpur (62%), Bareilly (61%), Darbhanga (59%), Sagar (49%), Patna (48%), Kota (41%) and Madhubani (only 3 orders, all returned). Together, these seven cities account for 16% of orders and have an RTO rate of 53%, compared with 24% across the other locations. Mumbai and Pune return only 18%.
- 2. Junk details predict returns, but they are rare. 140 orders had a very short or placeholder address such as ".", "xyz", "home" or "call me": 65% returned, against 26% for real addresses. 22 orders had fake names (asdf, test): 64% returned. Together they are only about 8% of orders, so they cannot be the whole fix.
- 3. New is risky, but "has returned before" is worse. First-time buyers return 30%. Repeat buyers with a clean history return only 14%. One earlier RTO means 35%, two means 76%.
- 4. Where orders come from matters. Meta ads orders return 31%, direct orders 23%. Orders above ₹1,999 return 41%, while ₹900–1,299 orders return only 22%.
- 5. The file itself is messy. 30 duplicate rows, status typed four different ways, 3 orders worth ₹0, 12 missing pincodes. Cleaning changed the numbers, so the same cleaning must happen on live Shopify data.

2. Recommended rules and what they do to last month

Each order gets a risk score (logistic regression on city, address quality, history, order value and source). The cut-offs live in `config.yaml`, so they can be changed without code. Results below are for September 2026, the most recent month in the data.

Risk	Action	Orders (Sept sample)	Actual RTO
Under 20%	Ship as-is	133	10%
20–45%	Nudge to prepaid (₹20–60 off)	144	32%
45–85%	WhatsApp confirmation first	51	51%
85% and above	Hold / cancel	1	100%

Result for 329 sample orders: an estimated ₹10,891 saved from avoided RTOs, against an estimated ₹4,027 cost, giving a net of ₹6,864. The 10% RTO in the "ship" bucket shows the score separates good orders from bad ones.

What the ₹4,027 is. It is an expected cost based on my assumptions, not a count of sales we actually lost. No genuine order was cancelled in September: the one held order was a real RTO. The ₹4,027 is made up of ₹1,993 from WhatsApp confirmations (an assumed 12% of customers whose orders were actually delivered would not reply or would decline, plus ₹1 per message) and ₹2,034 from prepaid offers (an assumed 25% of those customers switch to prepaid, costing ₹1,363 in discounts and ₹671 in prepaid-return risk). Most confirmed and nudged customers still receive their orders, so this is an estimated cost, not a count of lost sales. The ₹10,891 is also an estimate: it assumes 60% of would-be-RTO customers do not confirm, and 25% of nudged customers pay online.

Prepaid discount maths. For an average order of ₹649 (margin ₹402 when delivered, ₹169 cost per return) at 40% COD return risk:
• COD expected value = (60% × ₹402) − (40% × ₹169) = ₹173.60
• Prepaid expected value = (96% × ₹402) − (4% × ₹169) = ₹379.16 (prepaid returns only 4%)
• Maximum affordable discount = ₹379.16 − ₹173.60 = ₹205.56 (about ₹206)
We offer ₹60, which is about 9% of the order and under a third of the maximum, so most of the saving stays with us.

3. Operations vs Marketing, in rupees

All figures in this section cover the full six-month sample (April–September 2026, 1,984 orders), not a single month, and are not scaled to real volume. They use the same assumptions as above.

- Ops' plan (cancel everything at 40%+ risk): saves ₹37,776 of RTO cost but gives up an estimated ₹1,44,333 of margin on orders that were actually delivered, a net of −₹1,06,557. In our rules, cancelling paid off only at the very top of the range: in the 85%+ bucket, 9 orders saved ₹1,355 and cost ₹371.
- Marketing's plan (ship everything): net ₹0 by definition, and we keep absorbing about ₹78,000 a month in RTO cost (1,650 COD orders × 28% RTO × ₹169).
- Our plan: an estimated net of about ₹41,000 (₹66,366 saved, ₹25,414 cost, net ₹40,952) over the same six months. September's ₹6,864 is one month of this total.

Ops gets a hard stop for clearly bad orders (85%+ risk, or a junk address plus a fake name). Marketing gets a guard: a first-time customer is never cancelled on risk alone; they get a confirmation or a prepaid offer. Only 2 first-time customers were held in six months.

4. What could go wrong, and how we would notice

- My behaviour assumptions are guesses (60% of would-be-RTO customers do not confirm, 12% of genuine ones drop out, 25% accept prepaid). With pessimistic values (40%, 20%, 10%) the estimate falls to about ₹6,000 a month, and with very pessimistic values (30%, 30%, 5%) it turns negative, at about −₹9,500 a month. Check: weekly confirm-reply rate, drop-off rate and prepaid acceptance on the dashboard.
- Honest customers in the seven cities get extra friction. Check: sales per city before and after; if a city's orders fall more than its RTO, loosen that bucket.
- Scores go stale (new couriers, festivals, new ad campaigns). Check: actual RTO per bucket every week; "ship" should stay near 10–15%.
- Customers learn to ignore WhatsApp, or messages fail to deliver. Check: delivery and read rates.
- The AI misreads an address. The LLM never decides alone; it only adjusts a score, and failures fall back to rules.

5. If I had two more weeks

- Run a 10% random holdout (no action) for three weeks to measure the real catch, drop-off and acceptance rates.
- Connect the real WhatsApp API and a Shopify webhook so decisions happen live.
- Add courier-by-pincode serviceability and retrain monthly with the new outcomes.
- Test the LLM address check on 200 real orders and keep it only if it beats the rules.
- Add cohort tracking: do nudged or confirmed first-time buyers reorder?